

TaxaID: A Modular R Ecosystem for Bayesian Taxonomic Assignment

Project Overview and Purpose

Biologists increasingly measure biodiversity from sequence, sound, and image data, yet current pipelines can have high false-positive and false-negative rates for taxonomic assignment. TaxaID is a modular ecosystem of nine packages for the R programming language (R Core Team 2025; see Table 2, Software Requirements) that improve taxonomic assignment accuracy. A typical input is a table of candidate matches for each observation (sequence, image, or acoustic recording) previously obtained by querying a reference library. TaxaID can implement traditional score-threshold approaches (e.g., a fixed percent-identity cutoff paired with lowest-common-ancestor assignment, as in *galaxy-tool-lca*; Beentjes et al. 2019 – see Related Software, below), but its main advances are to:

1. screen for reference-database errors
2. detect and patch missing references
3. convert match scores to likelihoods
4. consider whether a taxon is apriori plausible at the sampling location
5. apply Bayes' Theorem to generate assignment probabilities from likelihoods and priors
6. make assignments transparent
7. use LLMs to improve assignments, assist with workflows, and review results
8. create shiny apps for clients to process their data

To apply Bayes' Theorem, TaxaID converts match scores to likelihoods, estimates spatially explicit occurrence-based priors (from GBIF and/or user data), and computes posterior probabilities of taxonomic identity. The ecosystem was designed with eDNA metabarcoding in mind, but image and acoustic analyses are possible when starting from a table of candidate matches.

The ecosystem supports three types of workflows, ranging from simple to comprehensive:

- **Traditional workflow** – Select a consensus taxon using score thresholds and/or a lowest common ancestor (covers many current approaches).
- **LLM-shortcut workflow** – Use large language models (LLMs) – Anthropic Claude (Anthropic PBC, San Francisco, California), Google Gemini (Google LLC, Mountain View, California), OpenAI (OpenAI OpCo, LLC, San Francisco, California), or local Ollama (Ollama, Palo Alto, California) – to rapidly estimate priors and generate consensus assignments.
- **Bayesian workflow** (the main purpose of TaxaID)– Train a likelihood model on reference data, build spatially explicit priors from GBIF (Global Biodiversity Information Facility; GBIF Secretariat, Copenhagen, Denmark) occurrences, and compute posteriors via Monte Carlo simulation.

These workflows converge at the same posterior consensus step, enabling direct comparison of model-based and LLM-based assignments.

Many TaxaID functions can use large language models (LLMs), though most have non-LLM alternatives. LLM integration requires an Application Programming Interface (API) key (see below).

Common Errors in Taxonomic Assignment

Automated classifiers for DNA, sound, and image data produce taxonomic assignments with systematic errors that are often difficult to detect. False presences are frequent in eDNA metabarcoding (Ficetola et al. 2015). In a replicated eDNA study from the California rocky intertidal, about 28% of the species detected could not be confirmed as occurring in the California Current System (Shea and Boehm 2024). Auto-classified

camera trap images carry error rates near 10% even at well-studied sites (Henrich et al. 2026), and acoustic classifier precision is highly sensitive to confidence threshold settings (Thompson et al. 2025; Fairbairn et al. 2025). Raw classifier scores mimic probabilities but are uncalibrated; 95% score does not mean 95% confidence (Dussert et al. 2025), and the same match percentage can be diagnostic for one taxon group but ambiguous for another (Pappalardo et al. 2025). These errors fall into three categories: false positives (FP; wrong taxon assigned, or overly confident in), false negatives (FN; correct taxon missed or underestimated), and combined errors where one taxon’s false positive is another’s false negative. FP/FN labels below mark which category each error mechanism produces.

Reference database quality **Reference mislabeling** (FP). Mislabeled sequences or images already present in the reference database can lead to false positives. *TaxaMatch* can check if accessions might be mislabeled so they can be removed before training models and generating consensus taxonomies. This generally follows three steps: *corroborate_references_locally()* is a relatively fast initial check for whether a reference has internal consistency within the data (i.e., a reference is similar to other same-named references). However, being overly cautious (removing correct references that don’t match well) biases the reference library (most suspicious references are correct). This merits a more extensive evaluation of suspect references. For sequences, *TaxaMatch* uses *evaluate_reference_accessions()* using a BLAST search to see if an accession matches taxa related to its label, and *review_flagged_accessions()* gives flagged/borderline accessions a third look from an LLM; after which *resolve_review_overrides()* double checks if flagged accessions should be kept or removed.

Missing reference redirect (FP + FN). The reference database itself is usually incomplete: when the true species has no entry in the reference library at all, its detections are assigned to the closest relative that does have an entry, a false positive for that relative and a false negative for the true species. Reference library gaps are geographically biased, systematically affecting some regions and taxa more than others (Marques et al. 2021). *TaxaLikely models the expected score profile of unreferenced taxa, and TaxaAssign identifies and names plausible missing species.*

Score interpretation **Overconfident species assignment** (FP + FN). Even when the correct species is present in the reference database, its raw match score is uncalibrated and taken at face value; a 100% match may still be ambiguous at species rank if competing candidates score nearly as well. Unlike the reference-gap problems above, this error arises purely from how scores are interpreted, not from what the reference database contains. *TaxaLikely’s calibrated likelihoods account for the fact that a high score with a small gap to alternatives has low species-level likelihood, regardless of the raw score.*

Overly strict thresholds (FN). Conservative score cutoffs discard correct assignments that fall just below arbitrary thresholds. *TaxaID’s probabilistic framework replaces binary thresholds with continuous likelihoods and posteriors.*

Ecological context **Defensive upranking** (FN). When multiple similar species produce near-identical scores, conventional systems uprank to genus to avoid a false positive, sacrificing species-level resolution. *Spatial priors from TaxaExpect can break ties; if only one candidate species is expected at the site, its posterior can support species-level assignment even when scores alone cannot. Dynamic Bayesian updating further sharpens priors within a sample after high-confidence detections support species presence.*

Ecologically implausible assignment (FP). A species is assigned that doesn’t plausibly occur at the sampling location, season, or habitat. *Spatially explicit priors from TaxaExpect down-weight implausible taxa. Dynamic updating within a sample reinforces ecologically consistent assignments.*

Field and lab artifacts **Contamination or artifact** (FP). Lab or field contamination, handler artifacts (camera traps), or equipment carryover introduces real detections of taxa not present in the environment. *TaxaFlag* can use information from blanks to remove contaminants from the source. Downstream, *TaxaFlag* uses LLM review to alert the user to candidates that look like contaminants or allochthonous transport from

outside the sampling area: eDNA carried by runoff or currents, sounds from playback devices or captive animals.

Ecosystem Packages

1. **TaxaTools** cleans and standardizes taxonomic names across backbones (GBIF; NCBI, the National Center for Biotechnology Information, U.S. National Library of Medicine, National Institutes of Health, Bethesda, Maryland; WoRMS, the World Register of Marine Species, Flanders Marine Institute (VLIZ), Ostend, Belgium; Catalogue of Life, hosted by Naturalis Biodiversity Center, Leiden, Netherlands) and provides a unified interface for calling LLMs from within the package.
2. **TaxaMatch** standardizes match tables from external tools (BLAST, the Basic Local Alignment Search Tool – Altschul et al. 1990, hosted by NCBI; camera-trap classifiers; acoustic detectors) into a common format, and screens reference accessions for mislabels via independent BLAST-based taxonomic congruence checking before the reference set is used for model training.
3. **TaxaLikely** converts match scores into calibrated likelihoods using a hierarchical Bayesian model trained on the reference library and auditing references for coverage gaps.
4. **TaxaFetch** acquires species occurrence records from GBIF, DataONE (Data Observation Network for Earth; University of New Mexico, Albuquerque, New Mexico), BioTIME (University of St Andrews, St Andrews, Scotland, United Kingdom), and published literature (including from PDFs).
5. **TaxaHabitat** classifies taxa into habitat categories using LLM-based biological consensus and flags spatial outliers.
6. **TaxaExpect** builds spatially explicit Bayesian priors by modeling expected species composition (each taxon’s relative share of the detections at a site) from observation records, incorporating habitat and spatial autocorrelation.
7. **TaxaAssign** computes posterior probabilities from likelihoods and priors, generates consensus taxonomy, and produces publication-ready reports.
8. **TaxaFlag** flags anomalous detections after assignment: contamination (lab/field blanks), handler artifacts (camera traps), and ecologically implausible assignments.
9. **TaxaWizard** interviews the user about their data and goals, then generates a complete R script, methods section, or Shiny application.

See the Readme.md file for each package for more details.

Dependency Chain (import order)

```
TaxaTools -> TaxaFetch -> TaxaHabitat -> TaxaExpect -> TaxaAssign -> TaxaFlag
TaxaMatch -> TaxaLikely -> TaxaAssign -> TaxaFlag
TaxaWizard (standalone; generates scripts that call the other packages)
```

The diagram below shows how information flows among packages: external data sources (parallelograms), external reference/occurrence databases (cylinders), the nine packages (rectangles), and final outputs (rounded terminals). **TaxaTools** is a cross-cutting utility (name cleaning, backbone conversion, LLM calls) rather than a single pipeline step, which is why it feeds several downstream packages directly rather than sitting in one place in the chain.

flowchart TD

```
%% Data sources
SEQ[/DNA sequences\n(FASTA / DADA2)/]
ACOUS[/Acoustic & camera-trap output\n(BirdNET / SpeciesNet / Animl)/]
PRECONSENSUS[/Pre-existing consensus\n(morphology / expert IDs)/]
LOCALREF[("Local reference DB\n(CRABS / FASTA)")]
NCBI[("NCBI / BOLD")]
```

```

GBIF[("GBIF / DataONE / iNaturalist")]

%% Packages
TaxaTools["TaxaTools"]
TaxaMatch["TaxaMatch"]
TaxaFetch["TaxaFetch"]
TaxaHabitat["TaxaHabitat"]
TaxaLikely["TaxaLikely"]
TaxaExpect["TaxaExpect"]
TaxaAssign["TaxaAssign"]
TaxaFlag["TaxaFlag"]
TaxaWizard["TaxaWizard"]

%% Outputs
OUT_CONSENSUS(["Consensus assignment\n+ methods report"])
OUT_REVIEWED(["Reviewed / flagged\ndetections"])
OUT_PRIORMAP(["Prior probability map"])
OUT_REFAUDIT(["Reference coverage audit"])

%% Sources into packages
SEQ --> TaxaMatch
ACOUS --> TaxaMatch
PRECONSENSUS --> TaxaTools
PRECONSENSUS --> TaxaFlag
LOCALREF --> TaxaLikely
NCBI --> TaxaLikely
GBIF --> TaxaFetch

%% Package-to-package data flow
TaxaMatch --> TaxaTools
TaxaTools --> TaxaFetch
TaxaTools --> TaxaLikely
TaxaTools --> TaxaExpect
TaxaTools --> TaxaAssign
TaxaFetch --> TaxaHabitat
TaxaHabitat --> TaxaExpect
TaxaLikely --> TaxaAssign
TaxaExpect --> TaxaAssign
TaxaMatch --> TaxaAssign
TaxaAssign --> TaxaFlag

%% Packages to outputs
TaxaAssign --> OUT_CONSENSUS
TaxaFlag --> OUT_REVIEWED
TaxaExpect --> OUT_PRIORMAP
TaxaLikely --> OUT_REFAUDIT

%% TaxaWizard sits outside the dependency chain
TaxaWizard -.->|generates workflow scripts calling| TaxaTools

classDef source fill:#e8f3ff,stroke:#5b8def,color:#1a3a6b;
classDef pkg fill:#fff7e6,stroke:#d99a2b,color:#5c3d05;
classDef output fill:#e9f9ee,stroke:#3fa85a,color:#1c4a2a;

```

```
classDef wizard fill:#f5e9ff,stroke:#9b59b6,color:#3d1f52,stroke-dasharray:4 3;

class SEQ,ACOUS,PRECONSENSUS,LOCALREF,NCBI,GBIF source;
class TaxaTools,TaxaMatch,TaxaFetch,TaxaHabitat,TaxaLikely,TaxaExpect,TaxaAssign,TaxaFlag pkg;
class OUT_CONSENSUS,OUT_REVIEWED,OUT_PRIORMAP,OUT_REFAUDIT output;
class TaxaWizard wizard;
```

TaxaWizard uses this flowchart to help users generate scripts (TaxaWizard/inst/graph/workflow_graph.json); see the *Pipeline Overview* section below for the finer-grained, object-level version.

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Licensing

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See LICENSE.md for details. TaxaExpect depends on glmmTMB (GPL ≥ 3); the TaxaExpect source code itself is CC0, but binary distributions that bundle glmmTMB may be subject to GPL terms.

Use of Large Language Models

TaxaID uses large language models in two distinct ways, and they should not be confused.

In the software. Several functions call an LLM as part of the analysis: TaxaHabitat assigns habitat categories by biological consensus, TaxaFlag reviews assignments for ecological plausibility, TaxaAssign offers an LLM-elicited alternative to the modelled prior, and TaxaWizard generates workflow scripts. All of these route through a common provider interface (`TaxaTools::call_api()`), so any supported provider can be used, and every one of them except TaxaWizard has a non-LLM alternative. Functions that make billed API calls cache their results to disk, so re-running a workflow does not pay for the same call twice.

In writing the code. The TaxaID source code was written with the assistance of Anthropic Claude models, used through Claude Code. Claude Sonnet did the bulk of the development; Claude Opus and Claude Fable were used more recently. Because this assistance was continuous rather than confined to particular functions, per-function annotation would imply a precision that does not exist, so this repository-level statement is the annotation. All of it was reviewed and tested before release: every package carries a `testthat` suite, and each was checked with `R CMD check` before release. The author reviewed the code and takes responsibility for it.

Related Software

Several R packages and standalone tools address taxonomic assignment from DNA barcoding data. TaxaID differs from these in its explicit separation of likelihood and prior, spatially explicit priors built from occurrence data, and multi-data-type support (DNA, image, acoustic).

Table 1. Comparison of TaxaID with related taxonomic assignment tools.

Tool	Approach	Posterior probabilities	Spatial priors	Unreferenced taxa	Multi-data-type
TaxaID	Generative Bayesian (MVN score + gap)	Yes (full distribution)	Yes (GBIF + habitat)	Yes (NCBI census + LLM)	Yes
PROTAX (Somervuo et al. 2016)	Bayesian with taxonomy-tree prior	Yes	No	Yes (tree-based)	No (DNA only)
BayesANT (Zito et al. 2023)	Bayesian nonparametric (kmer)	Yes	No	Yes (Pitman-Yor)	No (DNA only)
IdTaxa / DECIPHER (Murali et al. 2018)	Phylogenetic ML	Bootstrap confidence	No	No	No (DNA only)
insect (Wilkinson et al. 2018)	Profile HMM + classification tree	Akaike weights	No	No	No (DNA only)
SINTAX (Edgar 2016)	Kmer + bootstrap	Bootstrap confidence	No	No	No (DNA only)
RDP Classifier (Wang et al. 2007)	Naive Bayes (8-mer)	Bootstrap confidence	No	No	No (DNA only)
DADA2	Naive Bayes (kmer)	Bootstrap confidence	No	No	No (DNA only)
assignTaxonomy (Callahan et al. 2016)					
galaxy-tool-lca (Beentjes et al. 2019)	Score-threshold + LCA (deterministic)	No	No	No (upranked)	No (BLAST/DNA only)

TaxaID is complementary to several of these tools (and may load them for some purposes). DADA2 or OBITools handle upstream sequence processing; TaxaMatch ingests their output. DECIPHER is used internally by TaxaLikely for reference sequence alignment. The key innovation of TaxaID is that the same likelihood model can be combined with different spatial priors at different sites, and that the framework extends to image and acoustic data via TaxaMatch score standardization.

A large recent benchmark (Orsholm et al. 2026), spanning several tools listed in Table 1 (PROTAX, BayesANT) alongside similarity-, composition-, and phylogenetic-placement-based classifiers, confirms that no single algorithm dominates the reference-gap problem: phylogenetic placement (EPA-ng) performed best for arthropod COI barcodes, while composition-based classifiers (SINTAX, RDP-NBC, IDTAXA) performed best for fungal ITS, reflecting real differences in how alignable each marker is. This supports TaxaID’s own approach of treating reference-database completeness as a first-class, explicitly modeled problem rather than assuming one classification strategy generalizes across markers.

The galaxy-tool-lca code (Beentjes et al. 2019; <https://github.com/naturalis/galaxy-tool-lca>) is a widely used tool, written in Python (Python Software Foundation, Wilmington, Delaware), for LCA-based taxonomic assignment from BLAST results, particularly for freshwater macroinvertebrate eDNA and fungal Internal Transcribed Spacer (ITS) metabarcoding. Its core algorithm, which filters BLAST hits by identity, bitscore, and query coverage, then finds the lowest common ancestor among passing hits, is conceptually similar to TaxaID’s `score_consensus()`. A notable strength of galaxy-tool-lca is its explicit use of **query coverage** (the fraction of the query sequence that aligns to each reference hit) as a mandatory quality filter, alongside percent identity and bitscore. A 98% identity match that covers only half the amplicon is weaker evidence than one spanning the full amplicon, and coverage is already available as a BLAST output column (`qcovs`). TaxaID’s likelihood model currently uses score (percent identity) and the gap to the second-best candidate as its primary signals but does not incorporate alignment coverage. Users can partially address this upstream by filtering on coverage in TaxaMatch before passing match data to TaxaLikely; incorporating coverage as a third dimension in the likelihood model is a potential future enhancement. The analogous quality signal for acoustic reference data is the Xeno-canto (Xeno-canto Foundation, Netherlands, with support from Naturalis Biodiversity Center, Leiden; <https://xeno-canto.org/>) quality grade (A–E per recording); a dedicated helper for calibrating a quality-grade filtering threshold from it existed in TaxaLikely through 2026-09-09 and was archived after a real A/B test found the accuracy gain came with a real coverage cost (see TaxaLikely’s own README and CLAUDE.md) – a threshold can still be applied directly against the recorded quality-grade column before model training.

A parallel line of work addresses a different need: making metabarcoding output interpretable for managers and other non-specialist stakeholders through graphical interfaces. TaxonTableTools (Macher et al. 2021) provides platform-independent exploration and visualization of metabarcoding tables. The Pest Alert Tool (Zaiko et al. 2023) and BIOWATCH (Pearman et al. 2026) are targeted screening applications: both compare query sequences against a curated database of species of interest – non-indigenous, pathogenic, endangered, or commercially important taxa – and report which of those species appear in a dataset. BIOWATCH generalizes the Pest Alert Tool’s fixed Aotearoa-New Zealand marine list to user-defined species lists, markers, and regions, assembling custom BLAST databases with CRABS (Jeunen et al. 2023) and adding control- and replicate-based checks alongside spatio-temporal display of accumulated detections.

Several BIOWATCH steps have direct TaxaID counterparts: region-scoped species lists drawn from a map polygon (`TaxaTools::define_search_polygon()` with `TaxaFetch`), reference retrieval and in-silico amplicon trimming (`TaxaLikely::fetch_ncbi_reference_sequences()`, `fetch_bold_reference_sequences()`, `trim_to_amplicon()`; CRABS output itself is read by `TaxaLikely::read_crabs_output()`, database completeness auditing (`TaxaLikely::audit_barcode_coverage()`), control-based contaminant screening (`TaxaFlag::flag_contaminant()`), and watch-list surveillance (`TaxaFlag::flag_watch_candidates()`, `TaxaExpect::generate_invasive_watch_evidence()`). The substantive difference lies in how detection confidence is expressed. These tools resolve a detection into ordinal tiers – BIOWATCH labels a detection *Likely* when a single species uniquely holds the top percent identity within a bit-score window, and *Putative* when several species tie – which is a uniqueness heuristic applied to raw scores rather than a probability, and Pearman et al. (2026) identify probabilistic confidence metrics as future work. That is what TaxaID supplies: a calibrated likelihood combined with a spatially explicit prior yields a posterior probability for each candidate.

The two designs also differ in their exposure to reference-database gaps. A screening database restricted to target species plus their congeners guarantees that every query returns a best hit from within that restricted set, so a sequence from a taxon with no representation at all can still be reported against a target – the missing-reference redirect described above, and the reason TaxaID treats the absent-species response (H3) as an explicitly modeled hypothesis rather than a filtering threshold. Conversely, TaxaID has no counterpart to BIOWATCH’s long-term detection ledger, which accumulates detections with sampling metadata across years for spatio-temporal display. TaxaID’s scope ends at the assignment; analysis of the resulting detections is left to dedicated occupancy and trend tools.

Table 1b. Comparison of TaxaID (image path) with standalone image classifiers. TaxaID converts raw classifier confidence scores into calibrated likelihoods via a reference training set with known species identity, then multiplies those likelihoods by spatially explicit priors; the tools below produce the raw scores that

TaxaMatch and TaxaLikely process.

Tool	Taxa scope	Score output	Spatial priors	Unreferenced taxa	R access
TaxaID (image path)	Any (classifier-agnostic)	Calibrated likelihoods $\rightarrow$ Bayesian posteriors	Yes (GBIF + habitat)	Yes (coverage audit)	Native
animl / SpeciesNet (Tabak et al. 2019; Wildlife Insights*)	Camera trap mammals	Confidence (0–1), top-5; rollup ensemble	No	No	animl R package
iNaturalist computer vision	General wildlife (108,000+ taxa)	Softmax (0–1), top-10; genus/family fallback	Limited (app UI only)	No	rinat (indirect)
InsectNet (Chiranjeevi et al. 2025)	Insects (2,526 spp, 17 orders)	Conformal prediction sets; OOD energy score	No	Limited (OOD flag)	Web app only
Seek / iNaturalist mobile	General wildlife	Community consensus	No	No	None
Wildlife Insights*	Camera trap wildlife	Confidence (0–1), rollup ensemble	No	No	None (web platform)

* Wildlife Insights is a collaboration led by Conservation International (Arlington, Virginia) with Google LLC (Mountain View, California), the Wildlife Conservation Society (Bronx Zoo, Bronx, New York), WWF, the Smithsonian Institution, and others.

TaxaID is downstream of, not competing with, these classifiers. The key point is that raw confidence scores from neural networks are uncalibrated softmax outputs; a 90% confidence score does not mean a 90% chance the identification is correct. TaxaLikely addresses this by fitting a generative model to a labeled reference set with known species identity, capturing the full score distribution for correct matches (H1), wrong-species matches (H2), and absent-species responses (H3). **animl** (Conservation Technology Lab, San Diego Zoo Wildlife Alliance, San Diego, California) results are read directly by `TaxaMatch::read_animl_output()`; iNaturalist (a joint initiative of the California Academy of Sciences and the National Geographic Society, San Francisco, California) CV JSON output is read by `TaxaMatch::read_inaturalist_cv_output()`; SpeciesNet CLI (`google/cameratrapai`; Google LLC, Mountain View, California) batch predictions are read by `TaxaMatch::read_speciesnet_output()`. For acoustic and image data, TaxaLikely acts as a post-classifier calibration layer: classifier output is standardized to `match_df` by TaxaMatch, then `unreferenced_candidates()` + `assign_scores()` convert classifier confidence scores to likelihoods (no separate reference-building step required).

InsectNet (Chiranjeevi et al. 2025) is a notable recent advance for invertebrate specialists, achieving 96.4% top-1 accuracy across 2,526 species in 17 insect orders. Its standout methodological innovation is replacing point confidence scores with **conformal prediction sets**: rather than a single species estimate, the model returns a set of candidate species guaranteed to contain the true species with 97.5% probability, backed by an energy-based out-of-distribution score that flags images outside the training distribution. The conformal guarantee is conceptually related to TaxaID’s H1/H2/H3 framework; the true species is either in

the candidate set (H1/H2) or flagged as OOD (H3 analog). The outputs are not directly compatible with `train_likelihood_model()`, however, given the lack of access to InsectNet’s underlying softmax scores.

For contamination detection, the R package `decontam` (Davis et al. 2018) uses DNA concentration and prevalence to identify contaminants at the ASV level before taxonomic assignment. `TaxaFlag`’s proportion-based control comparison (`flag_contaminant()`) is typically applied the same way, before assignment, to remove likely contaminants when field or lab blanks are defined. However, `TaxaFlag` also runs a second, post-hoc pass after assignment: temporal proximity analysis, LLM expert review, and a combined view that rejoins the earlier contaminant flags against the final assignments.

Data and Hardware Requirements

- **Internet access** is required for GBIF queries, NCBI BLAST, and LLM API calls. Offline operation is possible when using cached data and local Ollama models.
- **NCBI rate limits:** a large remote-BLAST run (e.g. `TaxaMatch::evaluate_reference_accessions()` over hundreds of reference accessions) can hit NCBI’s own fair-use rate limiting or CPU-budget throttling partway through. `TaxaMatch` has some functions designed to make remote BLAST more robust. Importantly, it knows when to stop trying; `blast_sequences()`’s circuit breaker (`max_consecutive_batch_failures`) detects sustained failures and stops early rather than waiting out every remaining doomed batch; `evaluate_reference_accessions()` caches its results incrementally (`chunk_size`) so an interrupted or throttled run only loses whatever chunk was in flight, and reports a recommended pause before resuming the same call. `TaxaMatch::review_flagged_accessions()`’s LLM second-look review is real, billed API cost too, and caches on the same principle (`cache_dir`); re-running it only pays for accessions whose inputs genuinely changed since the last review, not the whole set again.
- **No specialized hardware** is required. All packages run on standard desktop hardware (macOS, Linux, or Windows) with R $\geq 4.1.0$.
- **Input data** varies by entry point:
 - *eDNA workflow*: DADA2 sequence table or FASTA file, plus sample metadata.
 - *Image/acoustic workflow*: match score table with taxonomic identifications.
 - *Name-only workflow*: a list of taxonomic names and site coordinates.

Software Requirements

Table 2. Software dependencies required for the TaxaID ecosystem.

Software	Version	OS bit	Reference
R	$\geq 4.1.0$	64	R Core Team. 2025. R: A Language and Environment for Statistical Computing. V.4.5.2. https://www.r-project.org .
Bioconductor (DECIPHER, Biostrings)	≥ 3.17	64	Gentleman et al. 2004. Bioconductor. https://www.bioconductor.org . Required only for <code>TaxaLikely::build_sequence_matrix()</code>

Software	Version	OS bit	Reference
rBLAST	≥ 0.99	64	Hahsler and Nagar. 2024. rBLAST. Bioconductor. https://doi.org/10.18129/B9.bioc.rBLAST . Optional; required only for local BLAST in TaxaMatch.

All other R package dependencies are declared in each package's DESCRIPTION file and will be installed automatically by `devtools::install()` or `install.packages()`.

API Keys

To access LLM tools, the user can use a locally installed LLM (Ollama) or an API key provided by an LLM service (set in `~/.Renviron`). A user could use a different LLM by modifying one of the existing LLM calls.

Key	Required By	How to Obtain
ANTHROPIC_API_KEY	TaxaTools (LLM calls)	https://console.anthropic.com/
GEMINI_API_KEY	TaxaTools (LLM calls)	https://aistudio.google.com/apikey (free tier)
OPENAI_API_KEY	TaxaTools (LLM calls)	https://platform.openai.com/ (paid)
OPENALEX_API_KEY	TaxaFetch (literature search)	https://openalex.org/settings/api (free)

No API key is needed for GBIF or NCBI queries. At least one LLM provider key is needed for the LLM-shortcut workflow, TaxaHabitat habitat assignment, and TaxaWizard.

Software Installation

```
# Install from GitHub
install.packages("remotes")

packages <- c("TaxaTools", "TaxaFetch", "TaxaHabitat", "TaxaMatch",
             "TaxaLikely", "TaxaExpect", "TaxaAssign", "TaxaFlag",
             "TaxaWizard")

for (pkg in packages) {
  remotes::install_github("DOI-USGS/TaxaID", subdir = pkg)
}

# Bioconductor dependency (needed for TaxaLikely reference matrix building)
if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")
BiocManager::install("DECIPHER")
```

After installation, load the packages:

```
library(TaxaTools)      # Name cleaning, LLM providers (auto-detects API keys)
library(TaxaFetch)      # Occurrence data acquisition
library(TaxaHabitat)    # Habitat assignment
library(TaxaMatch)      # Match standardization
library(TaxaLikely)     # Score-to-likelihood conversion
library(TaxaExpect)     # Spatially explicit priors
library(TaxaAssign)     # Posterior computation and consensus
library(TaxaFlag)       # Post-assignment quality flagging
library(TaxaWizard)     # Interactive workflow designer
```

Not all packages are needed for every workflow. At minimum, load TaxaTools (always required) plus the packages for your workflow.

Instructions for Using Software

Packages

Package	Purpose	License
TaxaTools	Name verification, cleaning, parsing, rank lookup, column standardization; LLM provider functions; GBIF backbone census	CC0
TaxaFetch	Occurrence data acquisition (GBIF, DataONE, PDF, literature search), source combination	CC0
TaxaHabitat	Habitat assignment via LLM biological consensus; spatial QAQC	CC0
TaxaMatch	Match standardization. Sequence input (DADA2/FASTA), BLAST search.	CC0
TaxaLikely	Convert match scores to likelihoods using hierarchical Bayesian model; reference QC	CC0
TaxaExpect	Spatially explicit Bayesian priors from occurrence and habitat data	CC0
TaxaAssign	Posterior computation, consensus taxonomy, report generation	CC0
TaxaFlag	Post-assignment anomalous detection flagging (contamination, transport, scope)	CC0
TaxaWizard	Conversational LLM workflow designer: guided interview to .R script, .md methods, or Shiny app	CC0

Pipeline Overview

The diagram below shows the full set of supported workflow paths. Every path converges at the assignment step; users can start from whichever inputs they have.

```

flowchart TD
  %% Field data inputs → match_df
  F1["seqtab / FASTA\n(eDNA / metabarcoding)"] -->|TaxaMatch| MD
  F2["BirdNET CSVs\n(acoustic)"] -->|read_birdnet_output| MD
  F3["camera trap classifier output\nAniml · iNat CV · Wildlife Insights"] -->|read_animl/inat/wi_out| MD
  MD(["match_df"])

  %% Reference library building → model_params (DNA only)
  R1["taxa names"] -->|"fetch_ncbi_reference_sequences\n+ audit_barcode_coverage"| RD
  R2["CRABS / local FASTA"] -->|"read_crabs_output\n+ read_reference_fasta"| RD
  RD(["reference_df"]) -->|build_sequence_matrix| RMAT(["DNA matrix"])
  RMAT -->|train_likelihood_model| MP(["model_params"])

  %% Priors
  TL["taxa + location"] -->|"TaxaFetch + TaxaHabitat\n+ TaxaExpect"| PR(["prior_df"])

  %% Likelihoods - three entry points
  MD & MP -->|"evaluate_likelihoods\n(DNA: uses model_params)"| LK
  MD -->|"assign_scores\n(acoustic / image: no model)"| LK
  CON["consensus_df\n(morphology / expert IDs)"] -->|"unreferenced_candidates\n+ assign_scores"| LK
  LK(["likelihoods"])

  %% Assignment
  LK -->|compute_posterior| PO
  PR -->|compute_posterior| PO
  PO(["posteriors"]) -->|"posterior_consensus\n+ generate_report"| OUT["consensus assignment\n+ methods report"]
  
```

TaxaWizard can generate a complete R script for any of the paths above via a guided interview: `workflow_create()`.

Which Entry Point Do I Need?

If you're not sure where to start, the fastest path is often the least code: `TaxaWizard::workflow_create()` interviews you about your data and goals and generates a complete, runnable R script for you (see Interactive Workflow Designer, below). If you'd rather find the right starting point yourself, match your data to a row below:

Your starting point	Start here
DADA2 sequence table or FASTA file (eDNA/metabarcoding)	<code>TaxaMatch::read_sequence_table()</code> / <code>TaxaMatch::blast_sequences()</code> – see <code>TaxaMatch/inst/workflow_fastq_to_match.R</code>
BirdNET CSV output (acoustic)	<code>TaxaMatch::read_birdnet_output()</code> – see <code>TaxaMatch/inst/workflow_image_acoustic.R</code>

Your starting point	Start here
Camera-trap classifier output (Animl / iNaturalist CV / SpeciesNet)	<code>TaxaMatch::read_animl_output()</code> / <code>read_inaturalist_cv_output()</code> / <code>read_speciesnet_output()</code>
A table of candidate matches you've already built (any data type)	Start at <code>TaxaLikely::evaluate_likelihoods()</code> (or <code>assign_scores()</code> for a no-training-data pathway)
Expert/morphological IDs with no match scores at all	<code>TaxaLikely::unreferenced_candidates()</code> + <code>assign_scores(score_type = "none")</code>
Just a list of taxon names and site coordinates (no observation data yet)	Fetch occurrences (e.g. <code>TaxaFetch::get_gbif_occurrences()</code>), then <code>TaxaExpect::estimate_kernel_priors()</code> – see <code>TaxaExpect/README.md</code> 's Quick Start.

For a worked example of a single pipeline stage, see that package's own `inst/workflows/*.R` scripts and its vignette – e.g. `TaxaLikely/inst/workflows/sequence_likelihood_workflow.R`, `TaxaMatch/inst/workflows/blast_sequences_workflow.R`, `TaxaAssign/inst/workflows/compute_posteriors_workflow.R`.

To build a complete workflow for a new site, start from `TaxaID_eDNA_Workflow_Template.R` (in the separate eDNA repository, at eDNA/PtConception/) – the template the three PtConception production workflows were built from.

To confirm your installation works, run one of the fast smoke tests. Each chains `evaluate_likelihoods()` -> `compute_posterior()` -> `posterior_consensus()` -> `add_slash_taxon()` against a small *real* fixture extracted from a completed production run, in a few seconds, with no network calls and no API key:

```
# from the repository root -- the fixture paths are relative to it
source("diagnostics/fast_workflows/run_fast_smoketest.R") # ~7 s
source("diagnostics/fast_workflows/run_greatlakes_fast_smoketest.R") # ~6 s
```

Both print warnings about hypotheses below `min_posterior` and about genus-level names in `plausible_taxa`. Those are expected: the fixtures use a flat placeholder prior, which is loudly labelled in each script. See `diagnostics/fast_workflows/README.md`.

Note: this package deliberately bundles no end-to-end runnable example. `inst/TaxaID_Workflow_Template_TEST.R` filled that role until 2026-09-15, when it was retired: its Section 5 called seven functions archived with the GLMM prior-fitting chain on 2026-09-09, so it had not been runnable for months while still receiving patches. A second in-package copy of a workflow is how that happened, so it was not replaced with another one – the canonical template is the single copy in the eDNA repository, and the smoke tests above are what verifies an install. See `archive_retired_workflow_template_2026_09_15/README.md`.

Getting Started

Each package includes a vignette with a worked example:

```
# Ecosystem overview (recommended starting point)
vignette("taxaid-ecosystem", package = "TaxaAssign")

# Individual package vignettes
vignette("name-cleaning", package = "TaxaTools")
vignette("data-acquisition", package = "TaxaFetch")
vignette("habitat-assignment", package = "TaxaHabitat")
vignette("match-standardization", package = "TaxaMatch")
```

```
vignette("score-to-likelihood", package = "TaxaLikely")
vignette("building-priors", package = "TaxaExpect")
vignette("taxonomic-assignment", package = "TaxaAssign")
vignette("quality-flagging", package = "TaxaFlag")
```

Workflow Scripts

Detailed, runnable workflow scripts are provided in each package's `inst/` or `inst/workflows/` directory:

Workflow	Package	Script
FASTQ to match data (eDNA)	TaxaMatch	inst/workflow_fastq_to_match.R
Image and acoustic identification	TaxaMatch	inst/workflow_image_acoustic.R
Fetch reference sequences (DNA)	TaxaLikely	inst/workflows/1_fetch_references_workflow.R
Flag reference errors	TaxaLikely	inst/workflows/2_flag_errors_workflow.R
Train likelihood model (DNA)	TaxaLikely	inst/workflows/3_train_model_workflow.R
Score to likelihood (image/acoustic)	TaxaLikely	inst/workflows/image_acoustic_likelihood_w
Score to likelihood (DNA)	TaxaLikely	inst/workflows/4_score_to_likelihood_workf
Audit reference coverage	TaxaLikely	inst/workflows/5_audit_coverage_workflow.R
No-score pathway (morphology/expert IDs)	TaxaLikely	inst/workflows/6_no_score_pathway_workflow
Demo: local reference library	TaxaLikely	inst/workflows/local_reference_demo.R
Fetch occurrences	TaxaFetch	inst/Merge_sources_workflow.R
Assign habitats	TaxaHabitat	inst/workflows/assign_habitat_workflow.R
Build priors	TaxaExpect	inst/TaxaExpect_workflow.R
LLM assignment	TaxaAssign	inst/TaxaAssign_llm_workflow.R
Bayesian assignment	TaxaAssign	inst/TaxaAssign_bayesian_workflow.R

For a step-by-step map of the full pipeline — inputs, outputs, and save conventions across all packages — see `ECOSYSTEM_WORKFLOW.md`.

High-Level Wrappers

For users who prefer a single-function interface:

```
library(TaxaAssign)

# Full Bayesian pipeline (~1 call, plus a saved model generated by TaxaLikely)
result <- run_bayesian_pipeline(match_df, model_params, taxaexpect_priors = taxaexpect_priors,
                                backbone_id = 11) # 11 = GBIF backbone; 4 = NCBI

# LLM-shortcut pipeline (~1 call)
result <- run_llm_pipeline(match_df, geographic_hint = "34.4 N, -119.8 W", habitat_scheme = "Marine",
                            backbone_id = 11) # 11 = GBIF backbone; 4 = NCBI
```

Interactive Workflow Designer

TaxaWizard provides a guided, conversational interface:

```
library(TaxaWizard)

# Opens a chat interface that interviews you about your data and goals,
# then generates a complete, runnable R script convertible to a shiny app.
workflow_create()
```

Troubleshooting

“No LLM provider configured” or an LLM call silently returns a uniform/degraded result. Confirm the relevant key (see API Keys, above) is set in `~/.Renviron`, not just your current shell session, then restart R. Provider auto-detection runs when a TaxaID package is attached with `library()` – calling functions only via `TaxaTools::call_api()` (fully namespaced, no `library()` call) skips it. If you’re running a script non-interactively (`Rscript`, not `RStudio`) and still see this after setting the key, pass a provider explicitly:

```
llm_fn <- function(prompt) TaxaTools::call_api(prompt, provider = "anthropic")
```

A recent bug fix or package update doesn’t seem to have taken effect. The most common cause is an R session that had the old version loaded before the update was installed. After any `remotes::install_github()` (or `devtools::install()` from source):

```
.rs.restartR()           # restart, clearing anything already loaded
library(TaxaTools)       # reload every package you use
packageDescription("TaxaTools")$Built # confirm this build is recent, not stale
```

If a workflow caches intermediate results to `.rds` checkpoint files (most of the worked-example scripts under each package’s `inst/` do this so long steps aren’t re-run unnecessarily), a fix to logic *downstream* of an existing checkpoint won’t show up until that checkpoint is deleted or the workflow is re-run with caching disabled – a raw-data fetch cache generally does not need clearing when only downstream filtering/statistical logic changed, but check the specific script’s own caching comments if a fix genuinely doesn’t seem to be taking effect.

GBIF or NCBI calls are slow, throttled, or fail partway through a large fetch. See the NCBI rate-limit note under Data and Hardware Requirements – functions that make many requests (`evaluate_reference_accessions()`, `blast_sequences()`, `download_gbif_occurrences()`) support `cache_dir`, so an interrupted run can resume from where it left off instead of restarting from scratch.

Getting help. If none of the above resolves it, please open an issue at <https://github.com/DOI-USGS/TaxaID/issues> with your R version, the exact error message, and a minimal reproducible example if possible – this helps other users hitting the same issue find the answer too, and keeps a public record other than a private email thread.

Data Outputs and Results

The TaxaID ecosystem produces outputs at each stage of the pipeline:

Occurrence Data and Habitat

- **Compiled occurrence records** from GBIF, DataONE, BioTIME, and literature extraction, standardized to Darwin Core columns (`stack_occurrences()`).

- **Habitat assignments** per taxon via weighted biological consensus across multiple habitat classification schemes (`assign_habitat_biological()`).
- **Spatial quality flags** identifying occurrences whose coordinates conflict with species habitat expectations (`flag_habitat_inconsistencies()`). These may be used to flag errant GBIF records.

Reference Library Assessment

- **Mislabel detection** identifying swapped or incorrectly labeled sequences in reference databases, via a free local-corroboration check followed by a BLAST-based screen (`TaxaMatch::corroborate_references_locally()`, `TaxaMatch::evaluate_reference_accessions()`). This can help users avoid basing assignments on reference errors.
- **Coverage audits** enumerating described species per genus and flagging taxa absent from the reference library (`audit_barcode_coverage()`, `audit_reference_coverage()`). This can help users understand limits in precision.
- **Model diagnostics** summarizing expected match percentages, score gaps, and per-species profiles (`interpret_model()`). This can help users understand how scores relate to precision.

Expected Composition and Priors

- **Site-centered kernel estimation** (current recommended path) predicting each taxon's expected share of the detections at a focal site from a distance-weighted (geo x optional-covariate) kernel over nearby occurrence records, with bandwidth chosen by leave-one-block-out composition prediction (`estimate_kernel_priors()`, `calibrate_kernel_bandwidth()`).
- **Beta priors** (alpha, beta) for every taxon at the focal site, including undetected diversity estimates (Good-Turing/Chao-anchored) for plausible but unobserved species (`generate_undetected_diversity()`).
- **KDE prior-field maps** – static or interactive Leaflet views of the estimator evaluated continuously across a lattice, not just at one site (`plot_theta_surface()`).

Taxonomic Assignment

- **Calibrated likelihoods** from a hierarchical Bayesian model trained on reference-vs-reference match scores, supporting three hypothesis types: known species (H1), unreferenced species (H2), and unreferenced genus (H3) (`evaluate_likelihoods()`).
- **Posterior probabilities** of taxonomic identity for each observation, with Monte Carlo uncertainty estimates (`compute_posterior()`).
- **Consensus taxonomy** assignments at the finest rank supported by the data, with confidence scores and consensus method labels (`posterior_consensus()`, `score_consensus()`).

Quality Control and Reporting

- **Quality flags** for anomalous detections: contamination scores from lab/field blanks, temporal handler proximity, LLM expert review (`flag_contaminant()`, `flag_handler()`, `review_assignments()`).
- **Publication-ready text** (Methods and Results sections) via template-based and LLM-assisted report generation, with per-package report sections that assemble into a unified document (`generate_report()`, `assemble_report()`).

Software Inventory

Package	Exported Functions	Test Files	Vignette
TaxaTools	48	23	Yes
TaxaFetch	32	25	Yes
TaxaHabitat	17	7	Yes
TaxaMatch	32	21	Yes
TaxaLikely	32	24	Yes
TaxaExpect	15	12	Yes
TaxaAssign	16	16	Yes
TaxaFlag	11	10	Yes
TaxaWizard	5	4	No

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